



MINING DEALER BEST PRACTICE

Tooling Setup for R&I of Steering Cylinders/Tie Rods in LOHT

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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April 2013
1301-4.03-1305

1.0 Introduction

The objective of this Best Practice is to outline the procedure and tooling required to effectively and safely remove/install steering cylinders and tie rods from Large Off-Highway Trucks (LOHT). A procedure currently exists in SIS for this process; however this Best Practice is designed to improve the tooling setup used to facilitate removal and installation. The benefits are primarily to ensure mechanic/fitter safety and reduce the physical exertion required in the removal/installation process.

2.0 Best Practice Description

Secure the steering cylinder/tie rod support frame by bolting it to the turntable of a common belly guard hoist. The turntable allows 360 degree movement of the support frame to easily align it with the steering cylinder or tie rod before they are disconnected. The belly guard hoist has a pneumatically operated motor to drive the wheels, eliminating the need for the operator to push and pull the hoist. The hydraulic lift system is contained within the hoist and is manually operated by the lever arm. These factors combine to produce an efficient, safe and an easy to execute method of steering cylinder and tie rod removal and installation in LOHT.

3.0 Implementation Steps

3.1 Hoist

Source an appropriately sized and rated hoist with turntable in accordance with the size of the steering cylinder or tie rod that is to be removed/installed from the truck. The hoist is hydraulically driven from a pneumatically operated motor, allowing the entire assembly to be moved easily. The flat surface of the hoist is raised or lowered by physically pumping the lever arm up and down, or via a separate lever (depending on model), to the desired level, as per a standard pallet jack. Refer to Sections 6 and Section 9 of this document for specifications and other details.

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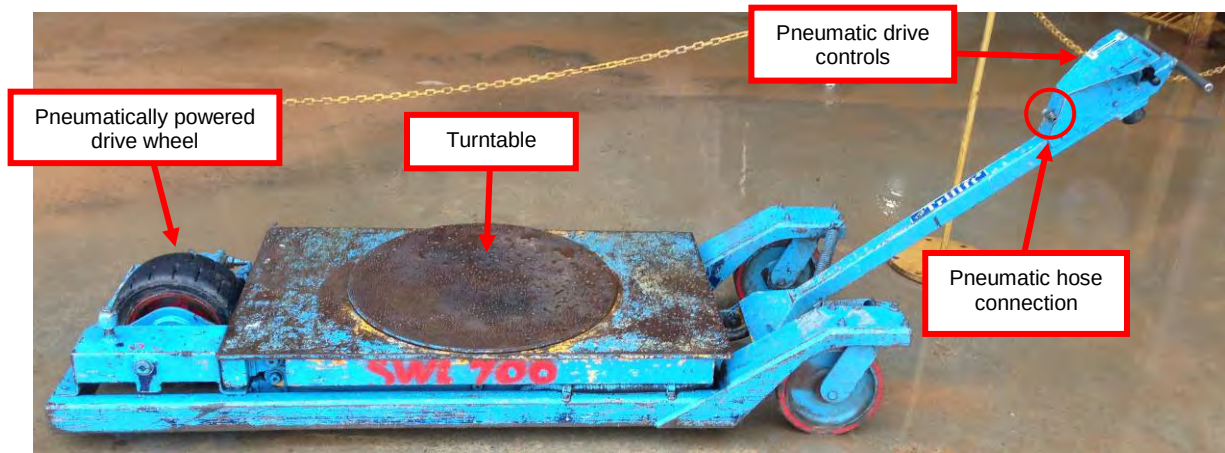


Figure 3.1 Belly guard hoist to be used as steering cylinder/tie rod hoist.

3.2 Support Frame

Ensure that the support frame conforms to the relevant local standard(s) for weight bearing requirements.

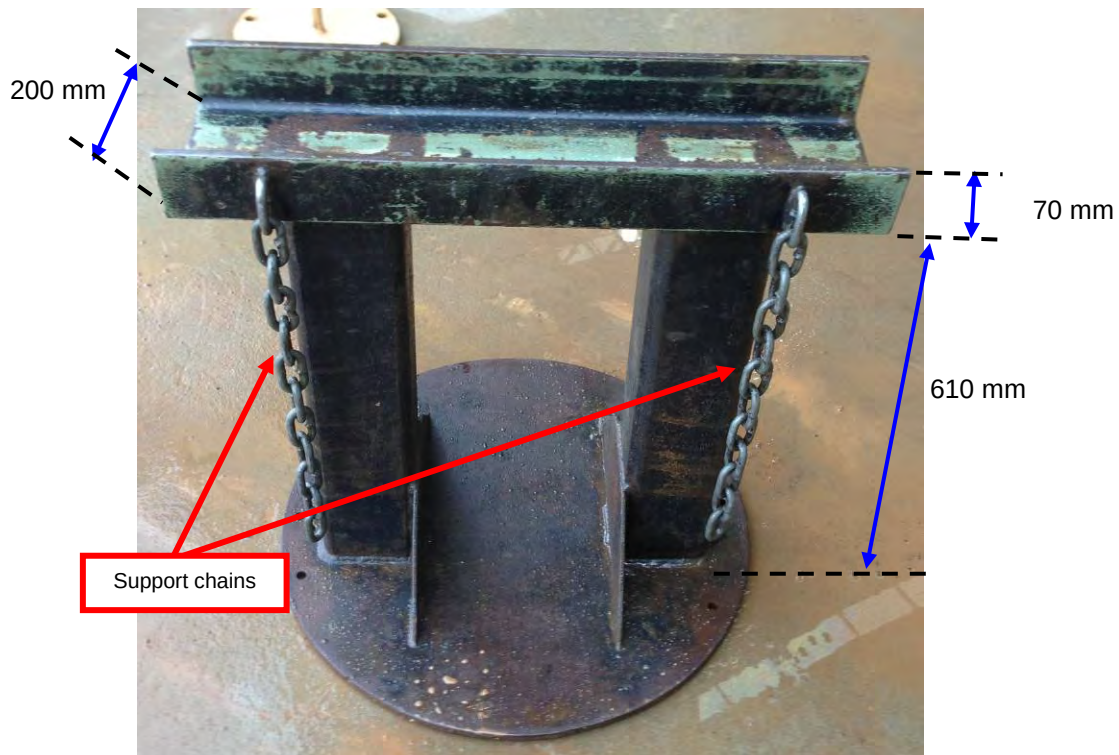


Figure 3.2 Support frame for steering cylinder/tie rod.

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3.3 Assembly

Secure the support frame to the hoist by bolting it to the turntable. Ensure that the type and number of bolts is in accordance with any relevant local standard to support the steering cylinder or tie rod. Note that the weight of a steering cylinder from a 797F is approximately 329 kg (725 lbs).

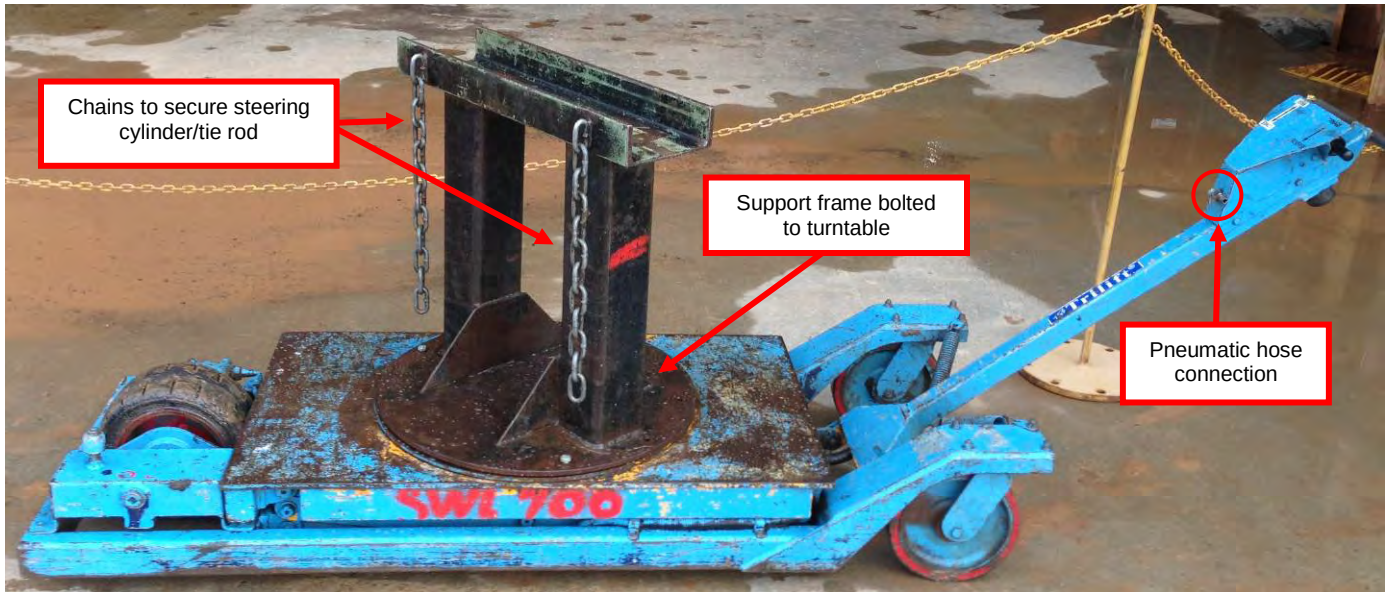


Figure 3.3 Support frame bolted to turntable on hoist.

3.4 Install & Remove

Install/remove the steering cylinder and tie rods using the hoist/support frame assembly in accordance with the Removal & Installation Instructions outlined in document KENR8579-04 located in SIS.

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Figure 3.4 Hoist raised and placed in position.



Figure 3.5 Support frame positioned under steering cylinder.

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Figure 3.6 Steering cylinder secured to support frame and jack lowered ready for removal from beneath the truck.

4.0 Benefits

- Assembly is robust and secure, reducing the likelihood of tooling failure or dropping of the steering cylinder or tie rod.
- Pneumatic drive motor eliminates manual pushing and pulling of the hoist, reducing operator fatigue.
- Increased efficiency by reducing component change out time.
- Since the hoist was originally designed to be used to remove and install the belly pan of various machines, it can also be used for this purpose.

5.0 Resources Required

- 1 x Steering cylinder/tie rod hoist with pneumatic motor drive control.
- 1 x Suitable support frame.
- 4-8 x Bolts (size and number as required) to secure support frame to the turntable on the hoist.
- Shifter/Pneumatic drill to secure support frame to the hoist.
- Pneumatic air supply to drive the hoist.

6.0 Supporting Attachments / References

- SIS document: KENR8579-04 (Steering Cylinder R&I – SMCS: 4303-010) or any other LOHT removal & installation document of a steering cylinder.
- Belly pan hoist sourced from Hedweld, Model number: TI10018.
(<http://www.hedweld.com.au/trilift/belly-pan-hoist-and-tooling/belly-pan-hoist>)

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7.0 Related Best Practices

N/A

8.0 Acknowledgements

This Best Practice was written by:

Adam Rieck – Equipment Management Consultant
Caterpillar Global Mining

The idea of creating and implementing this Best Practice belongs to:

Victor Lai – Equipment Management Consultant
Caterpillar Global Mining

Recognition:

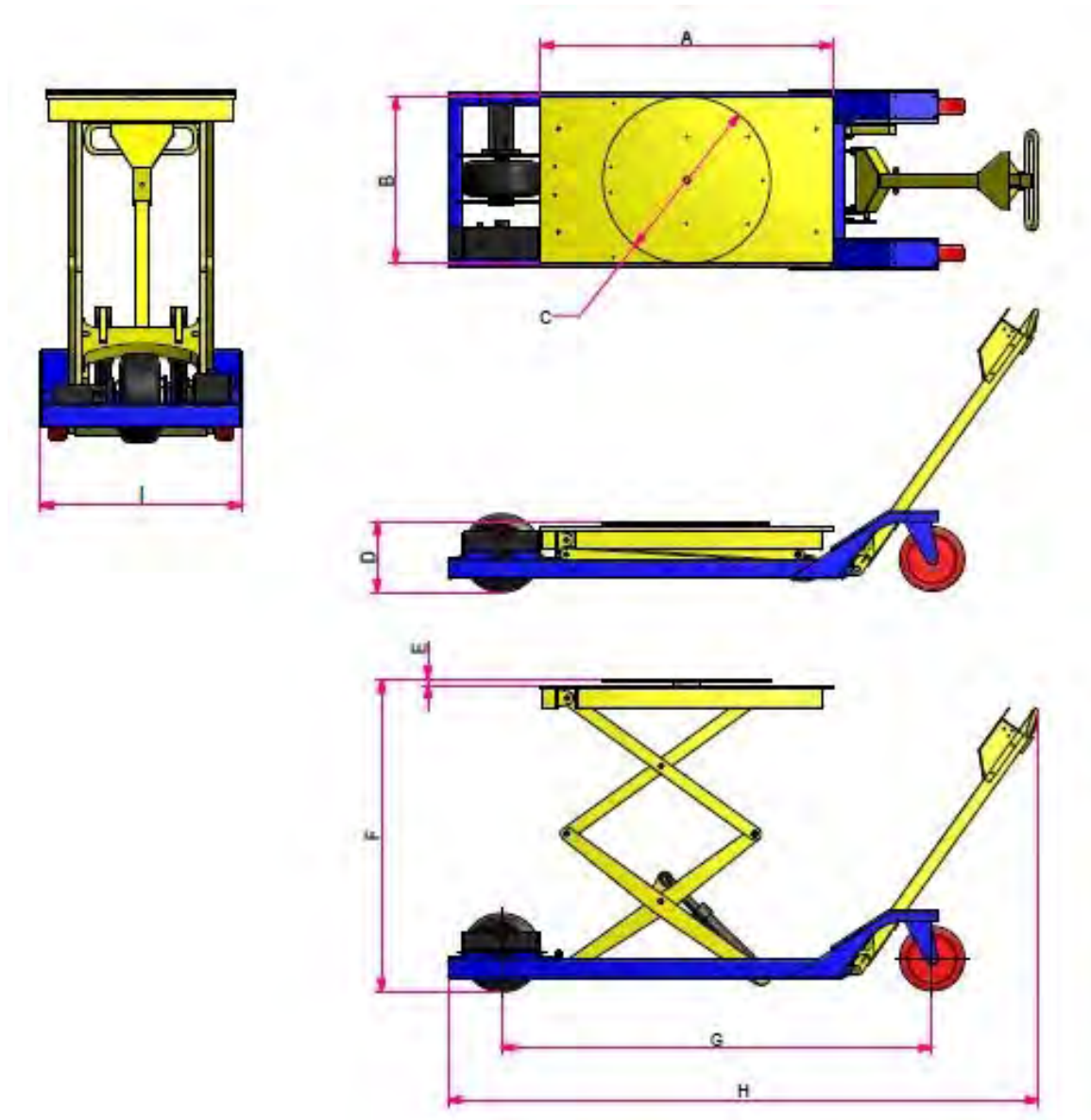
Lance Norbury – Workshop Supervisor
WesTrac WA – Boddington mine site

9.0 Appendix – Hoist Specifications

An example of a potential belly pan hoist sourced from Hedweld, Model number: TI10018.

Key Operating Data		
Safe Working Load	700 kgs	1,540 lbs.
Tare Weight	352 kgs	774 lbs.
Travel Speed	3.2 m/min	10.5 ft/min
Hydraulic Oil Tank Capacity	2.4L approx	0.63 gal
Motor	Air motor 1/3 hp -	
Nominal Air Pressure	6.9 bar	100 psi

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Key Dimensions

A	Table length	1136 mm	45 in
B	Table width	650 mm	25 in
C	Turntable diameter	650 mm	25 in
D	Lowered height	274 mm	11 in
E	Turntable height	22 mm	1 in
F	Raised height	1214 mm	48 in
G	Wheel centres	1671 mm	66 in
H	Overall length	2289 mm	90 in
I	Overall width	700 mm	28 in

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